DIGITAL ASSIGNMENT-6

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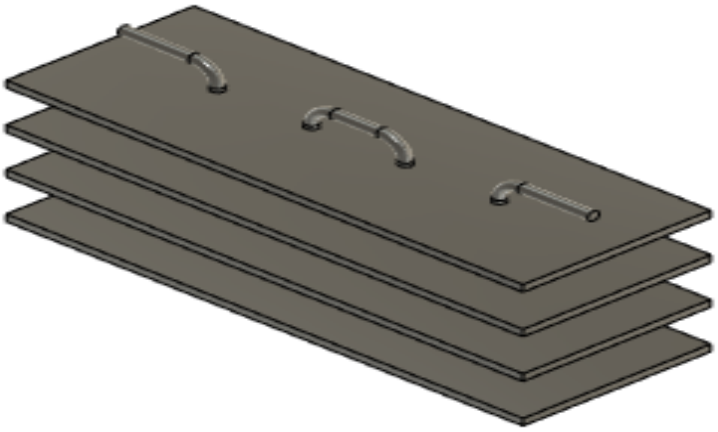
Aim:

To design a heat exchanger with fins, validate the thermal stress on the fins finally optimizethe fins dimensions to design an efficient heat exchanger

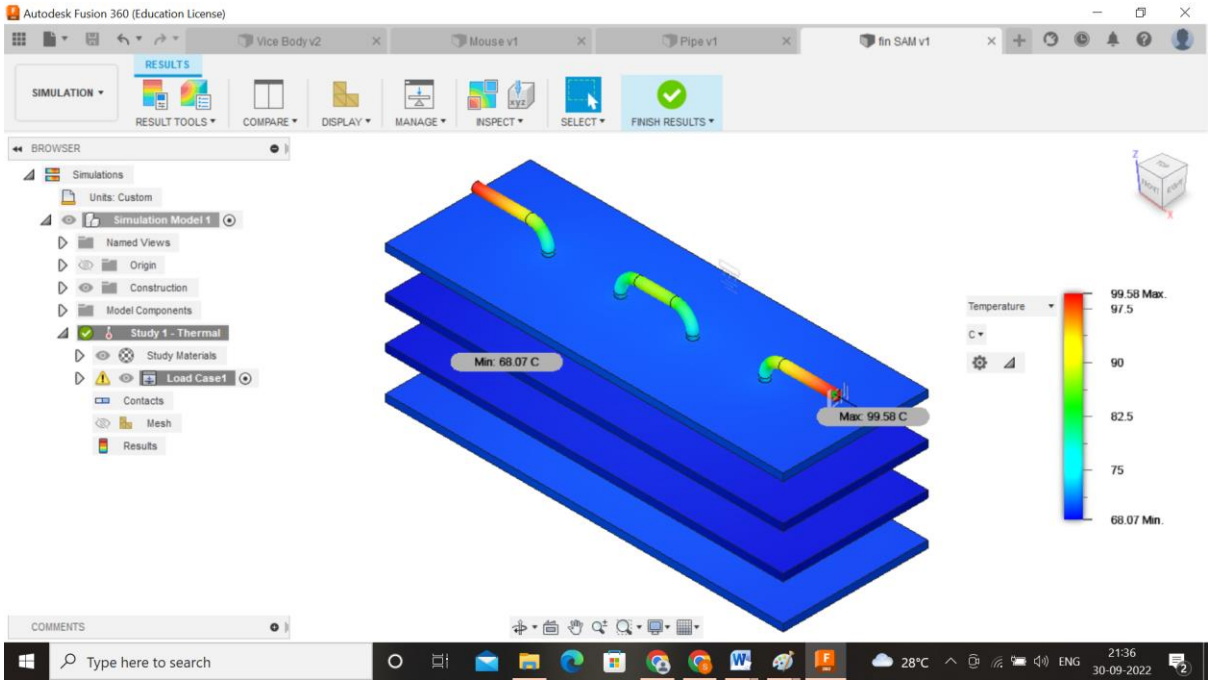
Procedure:

1. Design a exchanger with specified dimensions in a 2D workspace in the front view.

2. Use the fillet tool to design the fillets
3. With a specified dimensions draw a circle on the right view.
4. Use the swept command and design the exchanger pipe.
5. From the top view design, the fins and allocate specific dimensions and constrains.
6. Use the combine option and make cuts at the pipe cross-section
7. Now use the extrude command and extrude the fins make sure you give the off setdistance (offset distance should be more than the fillet radius).
8. Use the pattern tool and create multiple fins.
9. Now allocate materials (pipe-ss, finns-aluminium).
10. Now go to thermal analysis and specify the convention and load etc.
11. Create a mesh and simulate the design, check the results.
12. According to the results make changes to the fin dimension.



RESULT:



Link: https://a360.co/3SskQqG